A1 with generative AI: writing your automation landscape report in Claude

You may use a generative AI assistant to help produce your A1 report. This guide shows one way to do it with a free Claude account at claude.ai. It takes about three hours, most of it checking and writing. Claude drafts the textbook material, draws the diagrams in code and lays the report out as a Word document quickly. The marks for the gap analysis, the regulatory argument and the recommendations come from your own judgement about the Johannesburg encapsulation plant, so the guide keeps those parts in your hands.

Before you start

Sign in at claude.ai with a free account. Open Settings, then Capabilities, and turn on **Cloud code execution and file creation**. Claude needs it to read your PDFs, draw figures in code and hand you a real `.docx` file. You do not need to add any other skill: Claude's Word skill switches on by itself when you ask for a document.

If Claude stops and asks permission to run a command, click **Allow once**. The command runs in Claude's own sandbox in the cloud, not on your computer. A prompt that builds a file can take 5 to 15 minutes. Leave the tab open until Claude has finished and the file appears in the chat.

Start a new chat for this assignment and keep the whole job in that one chat, so Claude remembers what it has already built. A free account has a usage limit that resets every five hours. If you hit it, wait and carry on in the same chat.

Files to upload

Download these from the A1 assignment page on InnoSkills and attach all three to your first message, using the paperclip button:

1. `A01_automation_landscape_report.pdf`, your brief.
2. `A01_exemplar_cape_town_rooibos_report.pdf`, the worked exemplar report.
3. `A01_exemplar_cape_town_rooibos.pdf`, the marker's note on the exemplar.

The exemplar is set at a different plant (a rooibos drying and milling line in Cape Town), so its equipment, figures and findings would describe the wrong site. Use it to see the shape of a strong report: an executive summary that stands alone, every design choice tied to a finding, measured facts kept apart from the author's assumptions, and recommendations with an engineering column and a quality column.

Prompts

Copy each prompt into the chat in turn. Read the answer before you send the next one. Where a prompt has a square-bracketed gap, fill it in yourself.

1. Read the brief

```
I'm a learner on the DigiSkills programme. Attached are my A1 brief (PDF) and a worked exemplar report plus its marker's note from a different plant. Read the brief carefully. Then give me a checklist of every Key Requirement and every row of the marking scheme, with what a Distinction answer needs for each. Do not write anything yet. My report is about the Ubuntu Naturals Johannesburg encapsulation plant and its Moringa Daily capsule line, not the exemplar's Cape Town rooibos line.
```

2. Set out the current state

```
Here is what I know or have assumed about the systems on the Johannesburg Moringa Daily capsule line: [list each machine or system, its controller, any HMI, how its records are kept, and where your information comes from]. Put these in a table with the ISA-95 level each one sits at. Then list the levels that are empty or only partly covered. Mark every row that rests on my assumption rather than a source. Do not add equipment I have not named, and do not invent model numbers, quantities or outage figures.
```

Check the table against what you know of the plant. Which systems are on the floor, and which assumptions you state, are your decisions.

3. Plan the report

```
Propose a numbered outline for a 3,000-4,000 word report that covers every Key Requirement in the brief, with a word budget for each section. Include an executive summary, a contents list, a glossary and a references list. Mark three sections as mine to write: the gap analysis, the regulatory and ALCOA+ argument, and the recommendations. Do not draft any text yet.
```

4. Draft the comparison and protocol sections

```
Draft two sections for Johannesburg. First, the PLC vs. DCS comparison: a table with at least eight criteria (scalability, cost, GxP suitability, vendor lock-in, maintenance complexity, connectivity and two more), then a short paragraph stating my choice, which is [PLC / DCS / hybrid] because [your reasons]. Second, an evaluation of [three or four of Modbus, OPC UA, MQTT, WebSockets] against load-shedding resilience, edge buffering, GxP audit trail and limited bandwidth. Plain British English, short sentences. Where you state a fact a marker could check, name the kind of source it comes from; do not invent citations or figures.
```

Read both sections against your own notes and the lectures. Change anything you would not defend in front of the engineering team.

5. Draw the diagrams

```
Using Python, draw two figures for the report as PNG files at 300 dpi, sized to print 16 cm wide with no label smaller than 9 pt at that size. Figure 1: the ISA-95 pyramid for Johannesburg, showing what sits at each level and which levels are gaps, from the table in step 2. Figure 2: the network architecture from field devices up to MES/ERP, with the protocol at each boundary, the edge layer that carries the line through load-shedding, data flow arrows and a legend. Use colour that does not rely on red against green. Show me both figures.
```

Look at each figure at the size it will print. Ask Claude to fix any label that overlaps, clips or is too small.

6. Build the Word document

```
Build the report as A01_Automation_Landscape_Report_[YourSurname].docx. Arial 11 pt body text, numbered headings, a contents list, page numbers on every page, figures and tables captioned and numbered. Put in the sections drafted so far, the two figures, the glossary of every acronym used, and the references list. Leave the executive summary, the gap analysis, the regulatory and ALCOA+ section and the recommendations as headed empty sections, and leave no other section empty. Follow the exemplar's layout ideas but none of its content or wording.
```

7. Write the judgement sections yourself, then get feedback

Write the gap analysis, the regulatory and ALCOA+ argument and the recommendations in your own words, using your current-state table and your architecture. Then paste each draft into the chat:

```
Here is my draft [gap analysis / regulatory and ALCOA+ section / recommendations]. Mark it against the "[ISA-95 Pyramid Mapping / Regulatory and Data Integrity (ALCOA+) / Communication Quality]" row of the marking scheme in my brief. Tell me which band it reaches and what would lift it one band. Point out anything that contradicts the rest of the report. Do not rewrite it for me. My draft: [paste your draft here]
```

When you are happy with each draft, ask Claude to place it in its section. Write the executive summary last, from your own sections, and get feedback on it the same way.

Check before you submit

- Open the `.docx` in Word or LibreOffice Writer and confirm the headings are numbered, the contents list is up to date and every page has a number.
- Count the words in the body and confirm the total sits between 3,000 and 4,000. Say on the first page what the count includes.
- Open every reference and confirm it exists and says what you cite it for. Delete any you cannot find.
- Look at both figures at 100% zoom. Every label must be readable and every protocol named in the text must appear on the architecture diagram.
- Read the whole report for leftover exemplar content: Cape Town, rooibos, Mombasa, drying, milling or sieving should not appear anywhere.
- Export to PDF and name the file `A01_Automation_Landscape_Report_[YourSurname].pdf` with your own surname.

Add a short note at the end of the report saying which parts Claude built and which decisions and writing are yours. Claude can make mistakes. You are responsible for every number and every sentence in what you submit.